PAVINDRAN VELALAGAN

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EDUCATION

- **University of Peradeniya** *Jun 2023 - Present*
  BSc (Hons) in Computer Engineering | Current GPA: 3.31/4.00 Kandy, Sri Lanka
  - Societies: IEEE Computer Society, AeroGenZ Aviation Club, Perabeats Media Club

PROJECTS

- **Performance Interference Analysis in Heterogeneous CPU-GPU Systems (Research Project)** *Sep 2026 - Present*
  Linux, C/C++, Python, Bash, perf, PMU Counters, GPU Profiling, ARM, Nanopore Genomics
  - Investigating CPU, GPU memory interference in heterogeneous systems, focusing on performance degradation caused by shared resources such as memory bandwidth, cache, DRAM, interconnects, power, and thermal capacity.
  - Designing controlled co-execution experiments to compare isolated and concurrent CPU, GPU, and memory-intensive workloads and quantify their impact on execution time, throughput, and performance isolation.
  - Applying Linux performance-engineering tools and hardware performance counters to analyze CPU/GPU utilization, IPC, cache misses, memory bandwidth, stalls, memory traffic, and other indicators to identify the underlying causes of workload interference.
  - Exploring interference-aware resource-management techniques such as CPU affinity, workload co-scheduling, concurrency control, cgroups, and resource throttling, with planned validation using realistic Nanopore genomic-processing workloads.

- **Smart IV - End-to-End Infusion Monitoring & Control System (Group Project) [ ]** *Jan 2026 - Present*
  Rust/Tauri, ESP32, MQTT, C++, Hardware Design, IoT
  - Developed a closed-loop IV system using an IR drop counter and stepper motor mechanism to monitor and automatically regulate flow rate.
  - Integrated IoT connectivity via MQTT with mobile and centralized desktop applications for real-time monitoring and management.
  - Contributed to the Rust/Tauri desktop application and developed firmware for the edge device.
  - Received multiple national-level awards, runner up placements, and finalist recognitions.

PUBLICATIONS

J=JOURNAL, B=BOOK CHAPTER, R=UNDER REVIEW

[J.1] **Pavindran Velalagan**, et al. (2025). **ViT-RoT: Vision Transformer-Based Robust Framework for Tomato Leaf Disease Recognition**. *MDPI - AgriEngineering, Q1 Journal*. Published Jun 10, 2025.

[B.1] **Pavindran Velalagan**, Thiruthanigesan Kanagasabai, and Roshan G. Ragel (2026). **Ultra-Low-Power Video Coding for Wireless Capsule Endoscopy**. In *Fundamentals and Future Trends of Image and Video Coding*, IntechOpen. DOI: 10.5772/intechopen.1016872. (In press).

[R.1] **Pavindran Velalagan**, et al. (2026). **SLIF-Brinjal: An In-Field Leaf Dataset for Disease Recognition in Precision Agriculture**. Submitted to *Scientific Data (Nature Portfolio)*. Under peer review.

EXPERIENCE

- **Cogntix [ ]** *August 2025 - October 2025*
  ML Engineering Apprentice (Internship) Hybrid
  - Worked on text extraction techniques for complex, multi-column CV layouts.
  - Analyzed and tested methods for AI-based CV filtering to explore better candidate matching.
  - Contributed to the development of a document classification model to validate user uploads (CV vs. non-CV).
  - Gained hands-on experience with cloud deployment by building and hosting a CV parsing application using AWS (EC2, Route 53, Lightsail, Lambda).

Skills

- Programming Languages: Python, JavaScript, C, SQL
- Systems: Linux, Performance Profiling, PMU Counters, CPU/GPU Profiling
- Hardware & Embedded: Raspberry Pi, ESP32
- Tools & Cloud: AWS (EC2, Lambda, Lightsail), Git/GitHub, VS Code
- Data Science/ML: OpenCV, NumPy, Google Gemini API

Achievements

- **Winner - Best Innovative Product (Central Province) at SLASSCOM National Ingenuity Awards 2026** *Jun 2026*
  SLASSCOM (Sri Lanka Association for Software and Services Companies)
  - Awarded for Smart IV, a Medical IoT device developed as our 3rd Year Project.
- **1st Runner Up - NetX IoT Challenge 2026** *Sep 2026*
  University of Sri Jayewardenepura IEEE SB, IEEE ComSoc
  - 1st Runner Up in the IoT competition with the project Smart IV.
- **Finalist - Innov IoT Challenge 2026** *Sep 2026*
  SLIIT
  - Finalist in the IoT competition with the project Smart IV.
- **Finalist - SLIoT Challenge 2026** *Feb 2026*
  University of Moratuwa, SLT-MOBITEL, IESL
  - Finalist in the national IoT competition with the project Smart IV.
- **Finalist - Startup Spark 2024** *Oct 2024*
  Techno 2024, IESL Student Chapter - University of Moratuwa
  - Finalist for the Intelligent Traffic Monitoring System (ITMS) project, proposing an AI-driven solution with dynamic traffic signal control and emergency vehicle prioritization.
- **Finalist - Yarl Geek Challenge (Season 7)** *2018*
  Yarl IT Hub
  - Selected as a finalist for the Data Analytics Web Application project.

Volunteer Experience

- **Student Branch Ambassador** *Aug 2026 – Present*
  IEEEXtreme 20.0
  - Selected as a global Student Branch Ambassador representing the university for the 20th edition of IEEEXtreme, the world's largest 24-hour virtual programming competition.
  - Facilitate team registration, drive community engagement, and coordinate outreach efforts to encourage student participation.
- **Invigilator** *Jan 2026*
  ICPC (International Collegiate Programming Contest) Sri Lanka Peradeniya Regional Contest 2025/26
  - Served as an invigilator for the first-ever ICPC Regional programming contest hosted in Sri Lanka.
- **Casual Instructor** *Nov 2024 – Feb 2025*
  University of Peradeniya
  - Assisted undergraduates in CO1010 Programming for Engineers, covering Python fundamentals and practical problem-solving.
- **Webmaster** *Nov 2025 – Present*
  IEEE CS Student Chapter - UoP
  - Manage and maintain the chapter's web presence.
  - **Previous Role:** Marketing Team Member (Jul 2024 – Nov 2025). Contributed to graphic design and promotional campaigns for events like AlgoForge.
- **Marketing and Public Relations Coordinator** *Jan 2025 – Jan 2026*
  AerogenZ

Creative Experience

- **Founder & Photographer** *Sep 2023 - Present*
  Vivid Dreams
  - Established and operate a freelance photography brand specializing in portraiture and event photography.
- **Videography Team (Videographer / Video Editor / Crew)** *Dec 2023 - Jan 2026*
  Perabeats
  - Contributed as a Video Editor and Videographer (Dec 2023 - Jan 2025) and Crew Member (Jan 2025 - Jan 2026).